Predict at the point of application whether the lender will say No, so likely refusals can be pre-screened and routed faster

Business problem

- Every refused application still consumes **underwriting time** and gives the customer a poor experience.
- A refusal score lets the lender **pre-screen**, prioritise review with simple **Low / Medium / High** risk bands, and steer customers to a product they can get.
- Credit managers read the results in a **Power BI** dashboard, without needing Python.

Target: REFUSED

Approved	→ 0
Refused	→ 1
Canceled	removed
Unused offer	removed

Canceled and unused offers were decided by the **customer**, not the lender, so they are not part of the question.

Unit = one application (`SK_ID_PREV`). Client attributes joined on `SK_ID_CURR` from application_train + application_test (100% match).

Data and validation design

1.67M
applications in raw data

1.33M
decided (Approved / Refused)



Most recent 12 months (`DAYS_DECISION > −365`) held out as **out-of-time**. Older data split 70/15/15, **stratified**, seed 42. Preprocessing fitted on train only.

REFUSAL RATE & DATA QUALITY

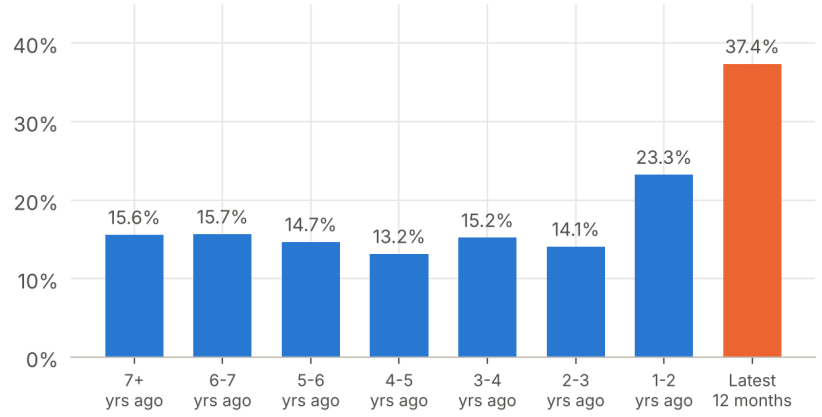
About 1 in 5 decided applications is refused, and the rate doubled in the latest year

21.9%

refusal rate (1.33M decided applications)

Moderate imbalance: 1 refusal per 3.6 approvals. Approving everyone would score **78.1% accuracy** and catch zero refusals, so models are ranked on **PR-AUC**. Handled with **class weights** + a **validation-tuned threshold**. No SMOKE: the minority class has 290k rows, and resampling distorts the probabilities behind the risk bands.

Refusal rate by year (years counted back from each client's latest application)



Leakage removed: information that only exists after the decision

- **13 post-decision columns**: disbursement and due dates (100% missing for refusals), `CODE_REJECT_REASON`, granted `AMT_CREDIT`, offer annuity and term, insurance-on-approval, last-application flags.
- **4 lender-assigned fields** found by ablation: portfolio, interest-yield group, down-payment rate, zero-amount flag. Removing any *one* changes PR-AUC by 0.001; removing all four costs **0.056**. The leak was encoded redundantly.
- **40,182 incomplete records** (blank field mandatory for the product, e.g. a cash loan with amount 0) are **98.7% refused**. They are excluded from modelling and shown as an *Incomplete* group.
- Client table is a snapshot at the **latest** application: age and job tenure re-computed **as of each decision**; future-dated fields not loaded.

Effect: the leaky model would have reported **PR-AUC 0.661**; the honest model scores **0.536**.

KEY EDA INSIGHTS

Product, credit history and channel drive refusals; demographics barely matter

4.2× **History:** no earlier refusal → 15.4% refused; after 1 → 33.3%; after 5+ → **64.8%**. Flag repeat applicants early and offer a smaller amount or different product.

3.2× **Product:** cash loans refused **34.7%** vs 10.7% for consumer (POS) loans. Fast-track POS, fuller review for cash and revolving.

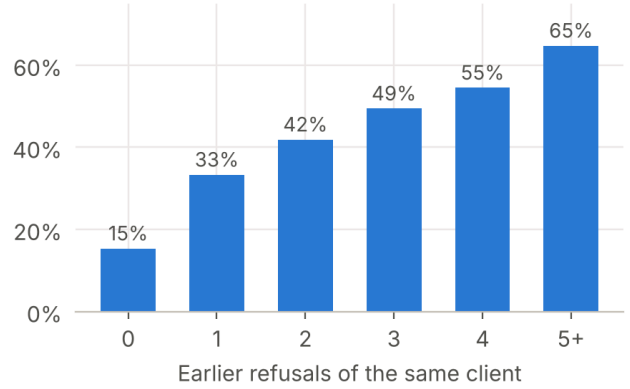
4.0× **Channel:** "AP+ (Cash loan)" **41.4%** vs "Stone" 10.4%. Credit & cash offices bring 33% of volume at 34.2%.

3.4× **Affordability:** asking for more than 4× annual income → **44.3%** refused vs 13.0% at 0.25–0.5× income.

2.1× **Acquisition:** walk-in applications **51.0%** vs 24.6% for cross-sell offers to existing customers.

2.2× **Trend:** latest 12 months **37.3%** vs 17.3% before: stricter lending and more cash loans. The model must be monitored for drift.

Refusal rate by number of earlier refusals



What matters less than expected

Refusal rate varies only **2.2 pts across age bands** (21.2–23.4%) and **6.3 pts across income bands** (18.6–24.9%). Also: new clients are refused 4.9% vs 28.4% for repeat clients, because repeaters include re-applications after a refusal.

Source: notebook 02 (all numbers computed in code)

MODEL COMPARISON & FINAL CHOICE

Interpretable baseline first; the stronger model must beat it on validation PR-AUC

Validation KPIs (complete records, 150,473 applications, refusal rate 15.9%)

Model	ROC-AUC	PR-AUC	Precision	Recall	F1	Flagged
No model (approve all)	0.500	0.159	–	0.0%	0.000	0%
Logistic Regression class-weighted · threshold 0.64	0.830	**0.536**	50.2%	56.7%	0.533	17.9%
LightGBM scale_pos_weight	*pending*	*pending*	*pending*	*pending*	*pending*	*pending*

Train vs validation gap < 0.01 AUC, so no overfitting. C tuned on validation (0.01 / 0.1 / 1: differences under 0.001). Same rows, features and splits for every model.

PENDING · LIGHTGBM

Final choice: **the model with the higher validation PR-AUC**. If LightGBM gains less than ~0.01, keep Logistic Regression, because it is simpler and its coefficients can be explained to credit managers.

Why these metrics

- **PR-AUC (main):** how well refusals are found across all thresholds; its baseline is the refusal rate, so it stays honest under imbalance.
- **ROC-AUC:** ranking quality, comparable across datasets.
- **Precision / recall / F1** at the chosen threshold: what the review team experiences.

Threshold: chosen on validation (max F1 = 0.64)

Threshold	Precision	Recall	Flagged
0.40 · catch more	31.4%	82.0%	41.5%
0.64 · chosen	**50.2%**	**56.7%**	**17.9%**
0.80 · fewer, surer flags	62.7%	38.6%	9.8%

Lower it if a missed refusal costs more than a wasted review.

Source: notebook 03 (Part A); LightGBM to be added in Part B

TEST VS OUT-OF-TIME RESULTS

Risk bands separate refusals clearly; OOT will test whether that holds on newer data

Final model: test (scored once) vs OOT

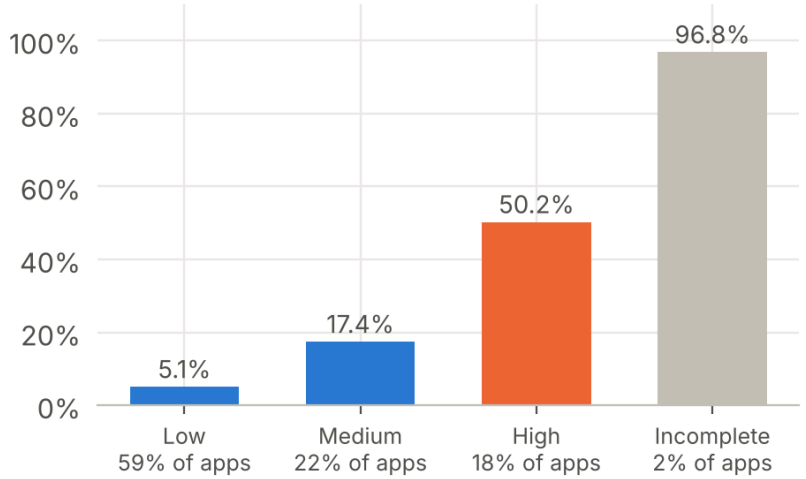
Split	Refusal rate	ROC-AUC	PR-AUC	Precision	Recall	F1
Validation	15.9%	0.830	0.536	50.2%	56.7%	0.533
Test	*pending*	*pending*	*pending*	*pending*	*pending*	*pending*
OOT	*pending*	*pending*	*pending*	*pending*	*pending*	*pending*

Validation row = Logistic Regression. Test and OOT are scored only for the final model, after selection.

What to expect on OOT, and why

- **Base rate shift:** refusal rate 17.3% → 37.3%. The lender is stricter, so **precision should rise and recall may fall** at a fixed threshold.
- **Mix shift:** more cash loans; incomplete records 1.7% → **7.4%** of applications.
- **Ranking (ROC-AUC)** usually holds better than threshold metrics; a PSI check will quantify the score drift.

Refusal rate by risk band · validation



High = probability ≥ threshold; **Medium** = above the 60th percentile; **Low** = the rest. The High band is refused **10×** as often as the Low band (50.2% vs 5.1%). Low-band applications are candidates for fast-track; High-band ones go to early review.

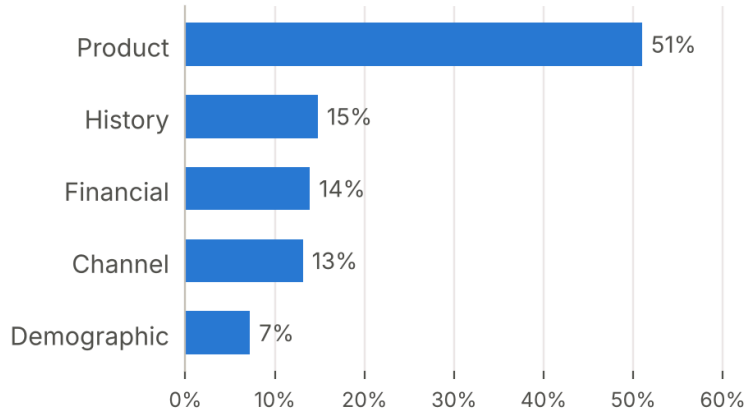
PENDING Replace with the final model's bands on test and OOT.

Source: notebook 03 · interim Logistic Regression

FEATURE IMPORTANCE & DASHBOARD

Product configuration and history drive the model; the dashboard puts it in managers' hands

Global importance by feature group (share of total)



Top features: product combination, channel, x-sell vs walk-in, goods price, client type (new/repeat), organisation type. This matches the EDA: product and history dominate and **demographics carry only 7%**.

Logistic Regression: sum of |standardised coefficients| per original feature. Cross-check with the final model's importance.

Power BI dashboard: star schema with 3 pages

Fact table of 1.33M applications; dimensions for client, product, channel, month, applicant profile and split. Measures include Applications, Refusal Rate, Refusal Lift, Avg Requested Amount and Avg Predicted Probability.

Screenshot: EDA overview

KPIs, refusal by product, channel and history, trend

Screenshot: Model performance

Test vs OOT, confusion matrix, risk bands

Screenshot: Predictions

Application table, probability, band, importance

Screenshot: Model view (star schema)

Source: notebook 03 · notebook 04 · refusal_dashboard.pbix

LIMITATIONS & NEXT STEPS

An honest, leakage-safe model, with known limits to monitor

Limitations

- **Client snapshot timing:** income and external scores describe the client at their *latest* application and cannot be back-dated, so older rows see slightly "future" client data.
- **Relative time:** dates are hidden. "Latest 12 months" is counted per client, not on a shared calendar, so the OOT window is approximate.
- **Drift:** the refusal rate doubled in the latest year, so thresholds and probability levels will need recalibration.
- **Same client in train and test:** the stratified split (as specified) lets one client appear in both, which can flatter test scores slightly.
- **Probabilities are not calibrated** (class weights push them up). Good for ranking and bands, not as literal chances.
- Lender policy is not observed directly. The model learns past decisions, including any past biases (gender deliberately excluded).

Improvements

- Stronger model (LightGBM) with light cross-validated tuning; **probability calibration** on validation.
- **Group split by client** as a robustness check.
- Monitoring: **PSI** on scores and key features, plus monthly refusal rate by band; retrain on a rolling window.
- Business-cost threshold: set the cut-off from the real cost of a wasted review vs a missed refusal.
- Add bureau and installment history *point-in-time* (only records dated before each application).

Bottom line: refusals are predictable at application time (validation ROC-AUC 0.83, PR-AUC 0.54 vs a 0.16 baseline), and simple risk bands separate a 5% from a 50% refusal group.